

Supplementary material

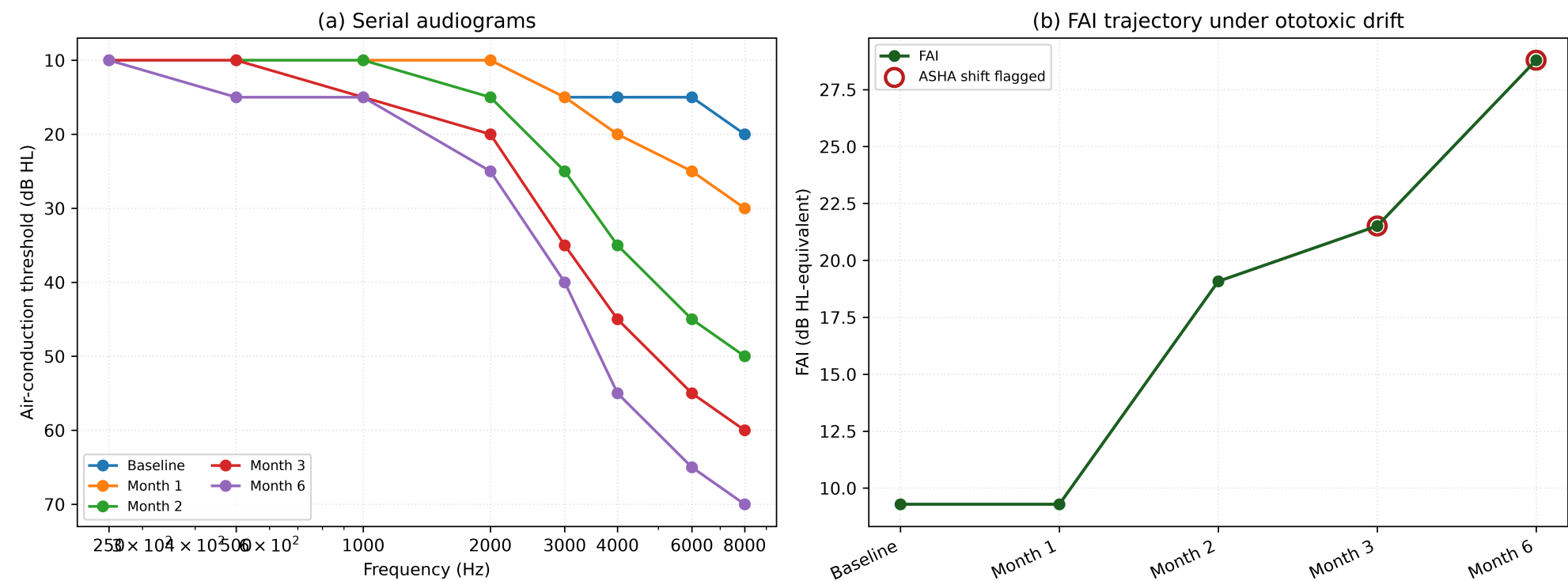
Ruspini partition parameters and the temporal tracking demonstration. Generated by scripts/make_supplementary.py from the fitted parameters and the live temporal module.

Table S1. Ruspini partition parameters

Category	P25 core (dB HL)	P75 core (dB HL)	Trapezoid [a, b, c, d]	Transition band to this category (dB HL)	Transition width (dB)	0.5 crossover (dB HL)	Calibrated label cut point (dB HL)
Normal	0.0	16.4	[-0.5, 0.0, 16.4, 30.0]	—	—	—	—
Mild	30.0	35.4	[16.4, 30.0, 35.4, 45.4]	16.4–35.4	19.0	23.2	23.8
Moderate	45.4	52.1	[35.4, 45.4, 52.1, 62.1]	35.4–52.1	16.7	40.4	48.9
Moderately severe	62.1	66.1	[52.1, 62.1, 66.1, 76.1]	52.1–66.1	14.0	57.1	63.8
Severe	76.1	85.0	[66.1, 76.1, 85.0, 95.9]	66.1–85.0	18.9	71.1	66.4
Profound	95.9	120.0	[85.0, 95.9, 120.0, 120.5]	85.0–120.0	35.0	90.5	95.0

The trapezoid [a, b, c, d] is the membership function for that category, with the core at [b, c] held at the empirical P25 and P75 and shoulders set so adjacent functions are complementary. Transition bands and 0.5 crossovers are geometric consequences of the parameters; the calibrated label cut points are fitted separately and are not the crossovers.

Figure S1. Temporal tracking under ototoxic drift



Serial audiograms (a) and the resulting FAI trajectory (b). Circles mark timepoints where the fuzzy-adapted ASHA shift criteria flag a significant change. Illustrative only: the cohort is repeated cross-sectional, so no longitudinal audiometric sub-cohort exists in these data.